

MAT1575, Classwork2, Fall2026

1.1 Approximating Areas (p.6-23)

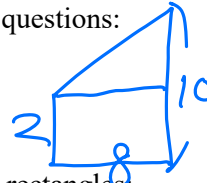
1. Example: Approximating Area of a Trapezoid under a given line.

Given a function $f(x) = x + 2$. Please finish the following questions:

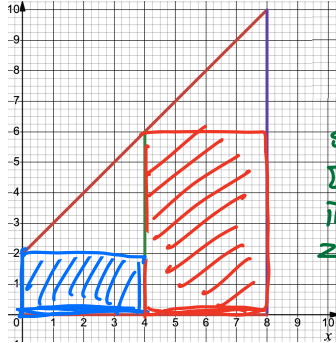
(a) Find the area between $y = 0$ and $f(x)$ for $0 < x < 8$.

Area $A = \frac{(2+10) \cdot 8}{2} = 48$

$$\begin{array}{r} x | 0 | 1 \\ y | 2 | 3 \end{array}$$

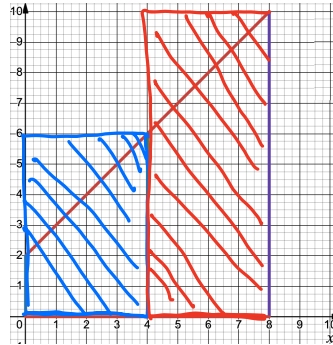


(b) Estimate the area in (a) by dividing it into 2 equal-width rectangles:



Left-endpoint approximation

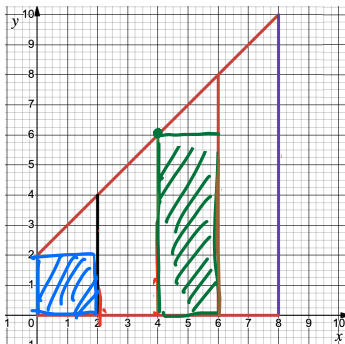
$L_2 = 2 \times 4 + 6 \times 4$
 $\text{splitting} = 8 + 24$
 $[0, 8] = 32$
 into 2 parts



Right-endpoint approximation

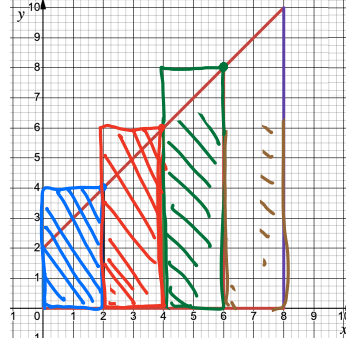
$R_2 = 6 \times 4 + 10 \times 4$
 $= 24 + 40$
 $= 64$

(c) Estimate the area in (a) by dividing it into 4 equal-width rectangles:



Left-endpoint approximation

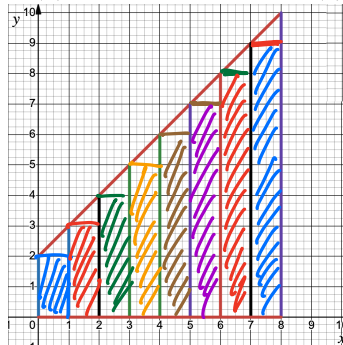
$L_4 = 2 \times 2 + 4 \times 2 + 6 \times 2 + 8 \times 2$
 $= 4 + 8 + 12 + 16$
 $= 40$



Right-endpoint approximation

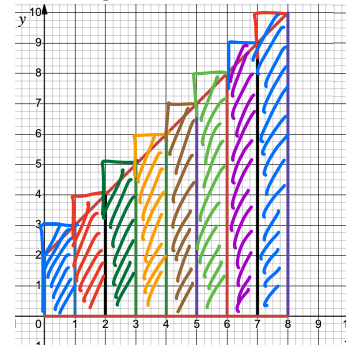
$R_4 = 4 \times 2 + 6 \times 2 + 8 \times 2 + 10 \times 2$
 $= 56$

(d) Estimate the area in (a) by dividing it into 8 equal-width rectangles:



Left-endpoint approximation

$L_8 = 2 \cdot 1 + 3 \cdot 1 + 4 \cdot 1 + 5 \cdot 1 + 6 \cdot 1 + 7 \cdot 1 + 8 \cdot 1 + 9 \cdot 1$
 $= 44$



Right-endpoint approximation

$R_8 = 3 \cdot 1 + 4 \cdot 1 + 5 \cdot 1 + 6 \cdot 1 + 7 \cdot 1 + 8 \cdot 1 + 9 \cdot 1 + 10 \cdot 1$
 $= 52$

(e) What is the relationship between $A, L_2, R_2, L_4, R_4, L_8, R_8$?

observation, if we have L_n (or R_n), what if $n \rightarrow \infty$

2. Definition: Partition

A set of points $P = \{x_i\}$ for $i = 1, 2, \dots, n$ with $a = x_0 < x_1 < x_2 < \dots < x_n = b$, which divides the interval $[a, b]$ into subintervals of the form $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$ is called a partition of $[a, b]$.

If the subintervals all have the same width, the set of points forms a regular partition of the interval.

Example in 1(b): $P = \{0, 4, 8\}$ it divides $[0, 8]$ into $[0, 4]$ and $[4, 8]$ which is a regular partition and width $\Delta x = 4$

Example in 1(c): $P = \{0, 2, 4, 6, 8\}$ it divides $[0, 8]$ into $[0, 2], [2, 4], [4, 6], [6, 8]$ which is a regular partition and width $\Delta x = 2$

and width $\Delta x =$

3. Left-endpoint approximation L_n and Right-endpoint approximation R_n

On each subinterval $[x_{i-1}, x_i]$ (for $i = 1, 2, \dots, n$), we construct a rectangle with width Δx and height h to calculate the area of this rectangle is _____. Adding the areas of all these rectangles, we get an _____.

(a) Left-endpoint approximation L_n

$h =$ _____ which is the function value at the _____ of the subinterval and

$$L_n =$$

$$=$$

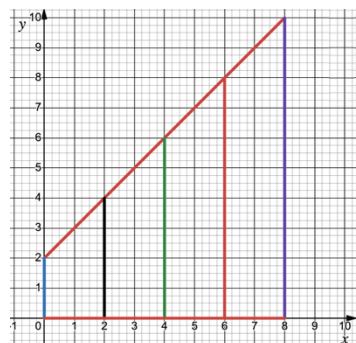
(b) Right-endpoint approximation R_n

$h =$ _____ which is the function value at the _____ of the subinterval and

$$R_n =$$

$$=$$

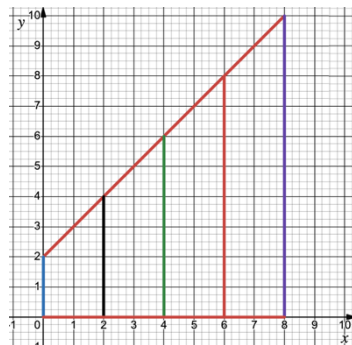
Example: $f(x) = x + 2$ with $P =$ _____



$n =$

$L_n =$

Example: $f(x) = x + 2$ with $P =$ _____



$n =$

$R_n =$

4. Review: Sums and Powers of Integers.

(a) The sum of n integers is given by $\sum_{i=1}^n i =$

(b) The sum of consecutive integers squared is given by $\sum_{i=1}^n i^2 =$

5. Review: Properties of Sigma Notation.

Let $a_1, a_2, \dots, a_n$ and $b_1, b_2, \dots, b_n$ represent two sequences of terms and let c be a constant.

$$(a) \sum_{i=1}^n c = nc. \quad (b) \sum_{i=1}^n ca_i = c \sum_{i=1}^n a_i \quad (c) \sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i \quad (d) \sum_{i=1}^n (a_i - b_i) = \sum_{i=1}^n a_i - \sum_{i=1}^n b_i$$

6. Example.

(a) Find the sum of the terms $(i - 3)^2$ for $i = 1, 2, \dots, 200$.

(b) Find the sum of the terms $j(j + 1)$ for $j = 0, 1, \dots, 9$.